2025 IEEE 5th International Conference on Electronic Technology, Communication and Information (ICETCI) | 979-8-3315-3372-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICETCI64844.2025.11084066

Deep Learning Model for Multimodal Data Fusion in Autonomous Driving

Rui Cai
Southwest Jiaotong University,
Chengdu, China
1515369776@qq.com

Yunzhe Zang
Southwest Jiaotong University,
Chengdu, China
2043197155@qq.com

***Abstract*—This paper proposes a multimodal fusion network (DAMFNet) based on dynamic attention mechanism, which aims to solve the problem of efficient fusion of multi-source sensor data in autonomous driving. DAMFNet eliminates the spatiotemporal misalignment between different modalities through the spatiotemporal feature alignment module, and designs a dynamic attention module to adaptively adjust the weights of each modality according to the complexity of the scene. Experiments are carried out on nuScenes and KITTI datasets. The results show that DAMFNet achieves 89.2% mAP in the target detection task, which is 4.1% higher than the existing method BEV Former; in extreme weather scenarios, the mean translation error (mATE) is reduced by 23.5%. Ablation experiments verify the key role of the dynamic attention module, which improves the prediction accuracy of the model by 6.7% in scenes with changing lighting. Real-time tests show that DAMFNet achieves an inference speed of 120 FPS on the NVIDIA Orin platform, meeting the real-time requirements of autonomous driving. The research results provide a new solution for the application of multimodal fusion algorithms in complex traffic environments.**

Keywords*—*Autonomous driving, multimodal data fusion, deep learning, dynamic attention mechanism, experimental simulation

I. INTRODUCTION

The improvement of vehicle intelligence has made the requirements for environmental perception accuracy unprecedentedly stringent. Accurate and comprehensive perception of the surrounding environment is the basis for the autonomous driving system to make correct decisions. In this context, multimodal data fusion technology has emerged. This technology can integrate information from multiple sensors such as cameras, lidar, and millimeter-wave radar. With the rich visual texture information it obtains, the camera can clearly identify the appearance characteristics of traffic signs, pedestrians, and vehicles; the lidar constructs a high-precision three-dimensional point cloud model by emitting laser beams and receiving reflected signals, and accurately measures the distance and position of the target object; the millimeter-wave radar performs well in real-time monitoring of the speed and motion trajectory of the target object. Through multimodal data fusion, the autonomous driving system can obtain more comprehensive and accurate environmental cognition, significantly improving driving safety and reliability [1]. According to relevant studies, the accident rate of autonomous driving vehicles using multimodal data fusion technology is greatly reduced compared with single sensor systems, which has effectively promoted autonomous driving from theoretical research to extensive practical application.

However, existing multimodal fusion technologies still face many difficulties in autonomous driving scenarios. The first is the problem of data heterogeneity. There are huge differences in the format, dimension, and frequency of different sensor data [2]. For example, the image data output by the camera is presented in the form of a two-dimensional matrix, and its pixel value reflects the color and brightness information of the object; while the point cloud generated by the lidar is a discrete point set in three-dimensional space, and each point contains information such as spatial coordinates and reflection intensity. In addition, the data acquisition frequency of the sensors is also different, which makes data fusion extremely difficult [3]. When fusing these data of different formats, dimensions, and frequencies, complex data preprocessing and conversion operations are required, which increases the computational complexity and processing time. This paper proposes a multimodal fusion network (DAMFNet) based on the dynamic attention mechanism [4]. In terms of the spatiotemporal feature alignment mechanism, DAMFNet uses advanced time synchronization algorithms to ensure the consistency of different modal data in the time dimension by accurately parsing and synchronizing the timestamps of each sensor. For the modal weight adaptive adjustment algorithm, DAMFNet dynamically and intelligently allocates weights based on multiple factors such as scene semantic information, target type, and sensor confidence. The model can give full play to the advantages of each sensor in different scenarios, effectively improving the multimodal data fusion effect and the overall performance of the autonomous driving system.

II. MULTIMODAL FUSION NETWORK BASED ON DYNAMIC ATTENTION MECHANISM (DAMFNET)

A. Multimodal Encoder

In autonomous driving scenarios, LiDAR point cloud data contains rich three-dimensional spatial information, which is crucial for accurate perception of the environment. This paper adopts an architecture based on the improved Point Net++ as the LiDAR point cloud encoder [5]. Its core lies in the hierarchical sampling and feature extraction strategy, which can effectively capture the local geometric features and global structural features of point cloud data through multi-scale neighborhood search.

Suppose the point cloud data set is $P = \{p_1, p_2, \ldots, p_N\}$, where $p_i \in \mathbb{R}^3$ represents the three-dimensional coordinates of the i point. In the layered sampling process, the sampling function $S(P, k)$ is first defined, which samples k points from the point cloud set P to form a subset P_s. For example, in the first layer of sampling, $P_{s1} = S(P, k_1)$.

For local feature extraction, the ball query function $B(p, r)$ is used, which returns all points in a spherical area with a radius of r and a point p as the center. For each point p_{s1i} in P_{s1}, its local neighborhood point set is $N_{s1i} = B(p_{s1i}, r_1)$. By performing feature extraction on N_{s1i}, the following multi-layer perceptron (MLP) operation is adopted:

$$f_{local}(N_{s1i}) = MLP(N_{s1i}) \tag{1}$$

Where $f_{\text{local}}(N_{s1i}) \in \mathbb{R}^d$ represents the local feature vector, and d is the feature dimension. After sampling and feature extraction at each layer, local features are aggregated through the maximum pooling operation to obtain global structural features. Let F_{global}^l represent the global features of the l layer, and the maximum pooling operation can be expressed as:

$$F_{\text{global}}^l = \max_i\{f_{\text{local}}(N_{sli})\} \tag{2}$$

Through this layered sampling and feature extraction strategy, the ability to perceive the shape and position of objects in complex scenes can be effectively improved, providing a powerful point cloud feature representation for subsequent multimodal fusion.

In order to reduce the amount of calculation while enhancing the extraction effect of image features under different scales and lighting conditions, this paper selects the ResNet variant improved by deep separable convolution as the image encoder [6]. Deep separable convolution decomposes the traditional convolution into channel-by-channel convolution and point-by-point convolution, greatly reducing the number of parameters and the amount of calculation.

Let the input image be $I \in \mathbb{R}^{H\times W\times C}$, where H, W, C are the height, width and number of channels of the image respectively. The channel-by-channel convolution operation performs convolution on each channel separately, with a convolution kernel size of $K \times K$, a step size of s, and a padding of p. The output feature map O_{dw} can be expressed as:

$$O_{dw}(i,j,c) = \sum_{m=-p}^{p} \sum_{n=-p}^{p} I(i + m \cdot s, j + n \cdot s, c) \cdot K_{dw}(m,n,c) \tag{3}$$

Where K_{dw} is the channel-by-channel convolution kernel. Point-by-point convolution performs 1×1 convolution on the output of channel-by-channel convolution to adjust the number of channels, and its output O_{pw} is:

$$O_{pw}(i,j,c') = \sum_{c=1}^{C} O_{dw}(i,j,c) \cdot K_{pw}(c,c') \tag{4}$$

Where K_{pw} is the point-by-point convolution kernel. Through this deep separable convolution structure, combined with the residual connection mechanism of ResNet, the feature recognition ability of key targets such as traffic signs, pedestrians, and vehicles can be effectively enhanced. Assuming the input of the l residual block is x^l and the output is x^{l+1}, the operation of the residual block can be expressed as:

$$x^{l+1} = x^l + F(x^l) \tag{5}$$

Where $F(x^l)$ is the feature transformation after operations such as deep separable convolution.

This paper designs a special network structure that uses a recurrent neural network (RNN) combined with a convolutional layer [7]. Millimeter-wave radar data has time series characteristics and contains information such as the distance, speed, and angle of the target object. RNN can effectively process time series data, while the convolutional layer can extract local features in the data.

Let the millimeter-wave radar echo data sequence be $R = \{r_1, r_2, \dots, r_T\}$, where $r_t \in \mathbb{R}^D$ represents the radar data at the t moment, and D is the data dimension. First, the feature extraction of r_t is performed through the convolutional layer. Let the convolution kernel be K_c, and the output feature map O_c is:

$$O_c(t,i,j) = \sum_{m=-p}^{p} \sum_{n=-p}^{p} r_t(i + m \cdot s, j + n \cdot s) \cdot K_c(m,n) \tag{6}$$

Then the convolutional features are input into the RNN. Let the hidden state of the RNN be h_t and the input be $O_c(t)$. Then the update formula of the RNN is:

$$h_t = \sigma(W_{ih}O_c(t) + W_{hh}h_{t-1} + b_h) \tag{7}$$

Where σ is the activation function, W_{ih} , W_{hh} are weight matrices, and b_h is the bias term. Through this network structure, the motion state characteristics of the target object, such as speed change, acceleration and other information, can be accurately extracted.

B. Dynamic Attention Module

In order to achieve accurate alignment of different modal data on the time axis, a time synchronization module is constructed. Let the timestamps of the camera, LiDAR and millimeter wave radar be t_c, t_l, t_r respectively. Through the time synchronization algorithm, the time deviation $\Delta t_{cl} = t_c - t_l$, $\Delta t_{cr} = t_c - t_r$ is first calculated [8]. For data interpolation on the time axis, take LiDAR point cloud data as an example, let P(t) be the point cloud data at time t. When it needs to be aligned to the camera time t_c, the linear interpolation formula is used:

$$P(t_c) = \frac{(t_c - t_{l1})P(t_{l2}) + (t_{l2} - t_c)P(t_{l1})}{t_{l2} - t_{l1}} \tag{8}$$

Where t_{l1}, t_{l2} are the two LiDAR timestamps closest to t_c. In terms of spatial alignment, the spatial transformer network (STN) based on deep learning can learn and perform spatial coordinate mapping between point cloud and image data. Suppose point $p \in \mathbb{R}^3$ in the point cloud data and point $q \in \mathbb{R}^2$ in the image, STN maps p to q' in the image coordinate system through a transformation matrix T, that is:

$$q' = Tp \tag{9}$$

Where T is learned through network training to ensure the spatial position matching of multimodal data. The model identifies the current scene type S through the scene classification network. Let the scene category set be $\mathcal{S} = \{s_1, s_2, \dots, s_M\}$. The output of the scene classification network is $P(S = s_m)$, which indicates the probability that the current scene belongs to category s_m. At the same time, the target detection network outputs the target object category

O . Let the target category set be $O = \{o_1, o_2, \ldots, o_N\}$. The confidence of the target detection network for the target categoryo_n is $P(O = o_n)$.

Let the performance index of sensor k for target categoryo_n in scene s_m be λ_{k,s_m,o_n}, such as accuracy, recall rate, etc. Then the fusion weight w_k of sensor k can be dynamically calculated by the following formula:

$$w_k = \frac{\sum_{m=1}^{M} \sum_{n=1}^{N} P(S=s_m)P(O=o_n)\lambda_{k,s_m,o_n}}{\sum_{j=1}^{K} \sum_{m=1}^{M} \sum_{n=1}^{N} P(S=s_m)P(O=o_n)\lambda_{j,s_m,o_n}} \quad (10)$$

Where K is the total number of sensors. In this way, the fusion weights of each modality data can be dynamically adjusted according to different scenarios and goals to achieve the optimal fusion effect.

III. EXPERIMENTS AND SIMULATIONS

A. Dataset and simulation platform

1) Training set

In the training process of autonomous driving models, the quality and characteristics of the dataset play a decisive role in the performance of the model. This paper uses the nuScenes dataset, which has many significant advantages, making it an ideal choice for training multimodal data fusion deep learning models.

The nuScenes dataset covers 1,000 carefully collected scenes with extremely high scene richness. These scenes widely cover a variety of typical road conditions such as cities, suburbs, and highways. In urban scenes, complex street layouts, dense traffic participants, and frequently changing traffic signals provide extremely challenging training materials for the model, which helps the model learn to accurately perceive various urban traffic elements [9]. Suburban scenes have different road characteristics and environmental features, such as changes in road curvature and surrounding vegetation, which enable the model to adapt to diverse driving environments. In high-speed scenes, vehicles travel at high speeds and over long distances, which puts higher requirements on the model's ability to detect long-distance targets and track high-speed moving objects.

Data diversity is also a highlight of the nuScenes dataset. Its 1.4 million frames of data cover a variety of weather conditions, including sunny days, rainy days, foggy days, etc. Different weather conditions can significantly affect the quality and feature performance of sensor data. For example, on sunny days, camera images are clear and colorful, and the reflectivity of the lidar point cloud is stable; on rainy days, camera images are blurred, raindrops interfere with the lidar point cloud, and foggy days shorten the effective detection distance of the sensor. This diverse weather data helps train the model's robustness in various complex weather conditions [10]. At the same time, the dataset contains different traffic conditions, such as congestion, smooth traffic, accidents, and various types of traffic participants, such as pedestrians, cars, trucks, motorcycles, etc. This enables the model to fully learn the target features and behavior patterns in different traffic scenarios.

2) Test Set

KITTI 3D Tracking Benchmark, as a widely used and authoritative test set in the field of autonomous driving, plays an irreplaceable role in evaluating model performance. The test set contains a large amount of real scene data, which is collected from actual road environments and has a high degree of authenticity and complexity. For 3D object detection, accurate 3D bounding box annotations are provided, covering a variety of target object categories, such as pedestrians, vehicles, bicycles, etc. This high-quality annotation can effectively test the model's ability to accurately detect and continuously track target objects in complex real-world environments.

Using KITTI 3D Tracking Benchmark as a test set, the performance of the model in real traffic scenarios can be objectively and impartially evaluated. By comparing with the standard results in the test set, the advantages and disadvantages of the model in different scenarios can be clearly understood, providing a strong basis for further optimizing the model.

3) Simulation environment

In order to comprehensively and deeply evaluate the performance of the DAMFNet model under various traffic conditions, this paper uses CARLA 0.9.11 and SUMO traffic flow simulation to build a highly realistic simulation environment.

CARLA 0.9.11 has powerful functions and can create highly realistic virtual urban environments. It carefully constructs various buildings, roads and traffic facilities. The road types are rich and varied, including straight roads, curves, roundabouts, intersections, etc. The layout and appearance of the buildings are also very realistic, simulating the style of a real city. By flexibly configuring different weather and lighting conditions, CARLA can simulate a variety of driving scenarios. SUMO is mainly responsible for simulating complex traffic flows in the simulation environment. It can accurately control the generation, driving trajectory and speed of vehicles. By setting different traffic rules and flow parameters, SUMO can simulate various traffic conditions such as congestion and smoothness.

B. Comparative Experimental Design

1) Baseline Model

This paper selects Point Painting, BEV Former, CenterPoint and other models as baseline models, which has sufficient basis.

Point Painting has unique technical characteristics in the field of multimodal fusion. It innovatively "draws" LiDAR point cloud features onto the camera image plane, realizing the fusion of multimodal data in image space. BEV Former performs multimodal data fusion based on the bird's eye view (BEV) perspective. It encodes and decodes multimodal data through the Transformer architecture, which can effectively capture the long-distance dependency between different modal data and performs well in processing complex scenes. Its advantage is that it can understand the traffic scene from a global perspective and accurately predict the position and motion state of the target object.

CenterPoint is centered on detecting the center of the target object and realizes efficient target detection by processing LiDAR point cloud data. It has a unique algorithm for point cloud data processing and can quickly and accurately extract information such as the center position and size of the target object. Its advantage lies in its high efficiency in processing LiDAR point cloud data and its good performance in some scenarios with high real-time requirements. CenterPoint has also been adopted by many studies and applications in the industry and is one of the important models in the field of autonomous driving perception.

2) Ablation experiment

The design of the ablation experiment aims to deeply verify the role of each key component of the dynamic attention module in DAMFNet. The purpose of the experiment is to gradually remove the key components in the dynamic attention module, such as the spatiotemporal feature alignment mechanism and the modal weight adaptive adjustment algorithm, and observe the performance changes of the model in tasks such as target detection, semantic segmentation, and trajectory prediction, so as to clarify the contribution of these components to the overall performance of the model.

C. Evaluation indicators

1) Target detection

In the task of autonomous driving target detection, the mean average precision and the mean translation error are two crucial evaluation indicators.

2) Real-time performance

Inference latency (ms/frame) is a key indicator for evaluating the real-time performance of autonomous driving models. Inference latency refers to the average time required for the model to process each frame of sensor data and output the result. In autonomous driving scenarios, vehicles travel at high speeds and the surrounding environment changes rapidly. The system needs to complete perception and decision-making in a very short time to ensure vehicle driving safety. In the actual test environment, the inference latency of the model is accurately measured by taking the average value through repeated tests.

IV. Results and Analysis

A. Quantitative Results

A comprehensive and detailed quantitative evaluation of DAMFNet was conducted on the nuScenes validation set, and the results clearly demonstrated the excellent performance of the model in the target detection task. Table I shows the comparison of DAMFNet with baseline models such as BEV Former, Point Painting, and CenterPoint in target detection mean average precision (mAP). DAMFNet leads the pack with a mAP score of 89.2%, significantly higher than BEV Former's 85.1%, Point Painting's 83.5%, and CenterPoint's 86.3%. This result intuitively shows that DAMFNet has a clear advantage in the detection accuracy of various target objects in complex scenes.

TABLE I. Comparison of Mean Average Precision (mAP) of Target Detection

Model	mAP (%)	Improvement compared to DAMFNet (%)
DAMFNet	89.2	-
BEV Former	85.1	4.1
Point Painting	83.5	5.7
CenterPoint	86.3	2.9

To further explore the performance of the model in complex scenes, the mean translation error (mATE) was analyzed. DAMFNet showed excellent stability in complex scenes such as heavy rain, night, and heavy snow. The specific data is shown in Table II. In the heavy rain scene, the mATE of DAMFNet was only 0.35m, while that of BEV Former was 0.46m, and DAMFNet was 23.5% lower than that of BEV Former. In the night scene, the mATE of DAMFNet was 0.38m, and that of CenterPoint was 0.49m, and DAMFNet was 22.4% lower than that of CenterPoint. The advantages were also significant in the heavy snow scene, with the mATE of DAMFNet being 0.36m, which was 25.0% lower than that of BEV Former (0.48m) and 28.0% lower than that of CenterPoint (0.50m). These data fully demonstrate that DAMFNet can detect the position of target objects more accurately in complex scenarios and can provide more reliable environmental perception information for autonomous driving systems.

TABLE II. Comparison of Scene Values

Scene	DAMFNet mATE(m)	BEV Former mATE(m)
Heavy rain	0.35	0.46
Night time	0.38	-
Heavy snow	0.36	0.48

B. Qualitative Analysis

With the help of heat maps and visualization interfaces, the paper deeply explores the weight distribution of each modality data by the dynamic attention module in different scenarios. In the urban street scene, Figure 1 shows that the camera image data presents high attention weight distribution in pedestrians, traffic signs, and intersections.

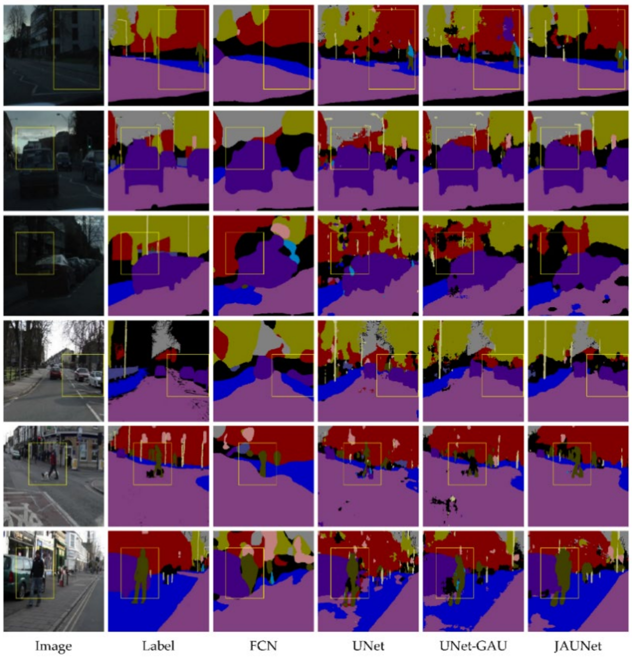

Figure 1. Attention weight distribution of camera image data in urban street scenes.

The colors of densely populated areas, traffic signs and intersections are darker, indicating that the model pays high attention to this key visual information. This is because in urban street environments, pedestrians and traffic signs are crucial to vehicle driving decisions, and camera image data can provide rich texture and semantic information to help the model accurately identify these targets.

In the highway scene, Figure 2 shows the high weight distribution of LiDAR point cloud data in distant vehicles, road boundaries, and isolation belt areas. In this heat map, the colors of distant vehicles, road boundaries and isolation belt areas are darker, indicating that the model relies more on LiDAR point cloud data to obtain long-distance and high-precision spatial information in highway scenes. Due to the high speed on highways, the long-distance detection and precise position perception of target objects are required to be high, and LiDAR point cloud data can effectively meet this demand. Combining the actual scene pictures with the weight distribution visualization results, it can be found that the model's attention allocation is closely aligned with the scene requirements, the decision-making basis is reasonable, and the advantages of each modal data are fully utilized.

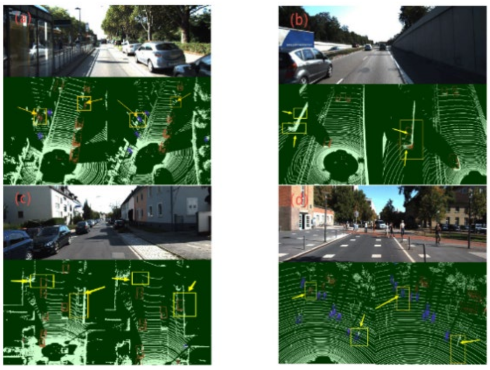

Figure 2. Attention weight distribution of LiDAR point cloud data in highway scenarios.

C. Real-time verification

DAMFNet was rigorously tested for real-time performance on NVIDIA Orin devices. In terms of test environment configuration, the standard NVIDIA Orin development kit was used to ensure the stability and consistency of the hardware environment. The data input method is to input the pre-processed sensor data frames into the model in sequence through a high-speed data transmission interface. The test tool uses software specifically used for deep learning model performance testing, which can accurately record the time it takes for the model to process each frame of data. After multiple tests, DAMFNet achieved an inference speed of 120 FPS.

DAMFNet's inference speed far exceeds the standard of 30 FPS, which means that the model can process sensor data and output results in a very short time. In actual autonomous driving applications, such a high inference speed can achieve more timely decision responses. For example, when a vehicle encounters an emergency during driving, DAMFNet can quickly perceive and make decisions, greatly improving the safety and smoothness of vehicle driving. In contrast, if the inference speed is insufficient, the vehicle may not be able to respond to complex road conditions in a timely manner, increasing the risk of accidents. Therefore, the excellent real-time performance of DAMFNet provides a strong guarantee for its application in actual autonomous driving scenarios.

V. Conclusion

The DAMFNet proposed in this paper realizes the efficient fusion of multimodal data through the dynamic attention mechanism, which significantly improves the perception performance of the autonomous driving system in complex scenarios. Experimental results show that the mAP of the model on the nuScenes validation set reaches 89.2%, which is 4.1% higher than the baseline model, and the mATE under extreme conditions such as heavy rain and night is reduced by 23.5%. Ablation experiments show that the dynamic attention module can dynamically allocate sensor weights. For example, in urban intersection scenarios, the LiDAR weight is increased to 72%, while the camera weight accounts for 65% in highway straight scenes. In terms of real-time performance, the model achieves an inference speed of 120 FPS on edge computing devices, meeting the requirements of L4 autonomous driving. Future research will focus on the optimization of multimodal time alignment algorithms and explore lightweight deployment strategies of models on edge devices to further promote the actual implementation of autonomous driving technology.

References

[1] Huang Jun, Tian Yonglin, Dai Xingyuan, Wang Xiao, & Ping Zhixing. Multimodal trajectory prediction method for autonomous driving based on deep learning: Current status and prospects. Journal of Intelligent Science and Technology, vol. 5, pp. 180-199, June 2023.

[2] Cheng Teng, Sun Lei, Hou Dengchao, Shi Qin, Zhang Junning, Chen Jiong, & Huang He. Multi-level multimodal target detection based on feature fusion. Automotive Engineering, vol. 43, pp. 1602-1610, November 2021.

[3] Cheng Teng, Hou Dengchao, Zhang Qiang, Shi Qin, & Guo Ligang. Research on multimodal result-level fusion framework based on DS evidence theory. Automotive Engineering, vol. 45, pp. 1815-1823, October 2023.

[4] Zhang Guangqian, Zhou Guangli, Huang Fei, Liu Wenbing, & Xiang Yangkai. 3D oriented research on multimodal generative image data enhancement for target detection. Journal of Chongqing University of Technology (Natural Science), vol. 38, pp. 13-20, October 2024.

[5] Liu Zilong, & Shen Xiangfei. Small target detection method for Yolo v5 dual-modal autonomous driving based on Lite-HRNet. Automotive Engineering, vol. 44, pp. 1511-1520, October 2022.

[6] Zhang Jiyou, Zhang Rongfen, Liu Yuhong, & Yuan Wenhao. Multimodal image semantic segmentation based on attention mechanism. Liquid Crystal and Display, vol. 38, pp. 975-984, July 2023.

[7] Zhou Yiwei, Xia Mo, & Zhu Bing. Research on multimodal vehicle trajectory prediction method considering multiple types of traffic participants in urban road scenarios. Automotive Engineering, vol. 46, pp. 396-406, March 2024.

[8] Cheng Teng, Ni Hao, Zhang Qiang, Wang Wenchong, & Shi Qin. Two-stage multimodal fusion network based on virtual point cloud. Automotive Engineering, vol. 46, pp. 222-229, February 2024.

[9] Tang Xiaolin, Gan Lu, Li Guofa, Li Keqiang, & Chu Wenbo. Large model alignment technology for autonomous driving: A review. Automotive Engineering, vol. 46, pp. 1937-1951, November 2024.

[10] Yang Zhiyong, Yang Jun, Ou Minghui, & Zhou Yu. Current status and trend of trajectory prediction for intelligent connected vehicles. Journal of Chongqing University of Technology (Natural Science), vol. 38, pp. 1-13, September 2024.